INCH-POUND

MIL-B-53083(ME)
19 August 1988

MILITARY SPECIFICATION

BAG, WATERPROOF, LOAD CARRYING;

SPECIAL OPERATIONS WATERPROOF BAG (SOWB)

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers, one type of special operation waterproof bag (SOWB) with automatic and manual buoyancy adjustments, watertight main compartment, exterior small non-waterproof pockets, and pack frame attachment loops and straps.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

V-T-285	- Thread, Polyester.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4220
DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Box, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-R-22368 - Repair Kit and Repair Kit Components, for Collapsible Fabric Tanks and Boats.
MIL-F-43834 - Frame, Field Pack, (Reverted), and Shelf, Cargo Support (Lightweight).

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.
FED-STD-595 - Colors.
FED-STD-601 - Rubber, Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attribute.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless other wise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents that is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 412 - Rubber Properties in Tension.
D 413 - Rubber Property-Adhesion to Flexible Substrate.
D 751 - Coated Fabrics.
D 822 - Light water exposure apparatus.
D 1149 - Rubber deterioration-Surface Ozone Cracking in a Chamber (Flat Specimen).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The SOWB shall consists of elastomeric coated fabric (see 3.6), water proof enclosure, sealed by a zipper, with 9 small exterior pockets, a compressed air cylinder and control valve, buoyancy adjustment valve, oral inflator, power inflator, and strap handles. The bag shall carry 60 pounds of equipment (40 pounds dry, 20 pounds wet). It shall remain waterproof at 0 pressure differential when submerged to a depth of 50 feet and when stored at a depth of 30 feet for 24 hours. The bag chassis, pockets, and all reinforcements shall be assembled of uncured rubber material as specified herein, and cured by vulcanizing. D-rings shall be provided for lifting, tiedown and attachment purposes, and to guide opening/closing of zipper. An air supply hose filling adapter, and repair kit shall be included with each bag.

3.2 First article. Unless otherwise specified (see 6.2), A sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of SOWB from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish SOWB confirming to the specification.

3.2.1 Sample selection plan. Prior to initial product, the contractor shall have a sample selection plan. The plan shall be designed for use in the quality conformance testing of section 4 and shall provide the Government a method of selecting production seams, bonded fitting, coated fabric and handle samples for destructive tests. The samples of cloth shall be cut and selected at random in accordance with 4.4.1. The test specimens of seams, bonded fittings, coated fabrics and handles shall be provided in sufficient quantity for testing as specified herein. The sample selection plan shall be made available, upon request, to the contracting officer or Government representative on site of manufacturer. The test specimens may be made specifically for testing but will be certified as using the same material and procedures as those used for production items.

3.2.2 Bags for testing. SOWBs selected for samples are in addition to the quantity required for delivery and shall remain the property of the contractor after completion of tests. All SOWBs selected for the purpose of providing samples for destructive tests are complete with attached pockets, straps and fittings.

3.2.3 Additional first article samples. Unless otherwise specified (see 6.2) additional first article samples shall be provided to the U.S. Army Belvoir

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

Research, Development and Engineering Center, ATTN: STRBE-VU, Fort Belvoir, VA. 22060-5606, one week prior to commencement of first article testing. Samples shall include all samples required to run sample tests in table I, II, III plus one square foot of each compound used. All samples shall be marked with appropriate contract number.

3.3 Process data. Prior to initial product, the contractor shall make available to the Government representatives, on site of manufacture, a description of materials, documented work instructions, processing and fabricating procedures which will be used by the contractor to assure that each production SOWB is a duplicate of the first article bag. Change of materials, components, design, procedures, or sources of supply from those initially selected shall be subject to the approval of the contracting officer.

3.4 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4.1 Material deterioration prevention and control. The SOWB shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the SOWBs may be exposed.

3.4.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish, or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4.1.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the bags may be newly fabricated from recovered materials to the maximum extent practicable, provided the bags produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the bags.

3.4.1.4 Material documentation. Each bag shall be issued a serial number. (see 3.15). Documentation of date and location of material manufacture shall be available for inspection. In order to ensure proper assembly before vulcanization, the rubber materials used in the manufacture of the bag may not be older than 30 days at the date of manufacture of the bag. The contractor shall maintain records of the material manufacture date and bag fabrication date for not less than 5 years from date of delivery.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.4.2 Nylon cloth. The nylon cloth (see 3.4.4 and V-T-285) of the coated fabric shall be high-tenacity, and improved heat and light resistant. The nylon cloth shall be free from any imperfection affecting strength, coating adhesion, or coating thickness conforming to V-T-285.

3.4.2.1 Weathering resistance. When tested as specified in 4.5.2.5.10, the nylon cloth shall have retained breaking strength that is not less than 45 percent of its initial breaking strength warp and fill pounds (lbs) minimum (min.).

3.4.3 Coating compounds. Each and every compound used for the formulation of elastomeric coatings, tie gum, diffusion barrier (if used), and any other materials applied to the coated fabric shall conform to table I. The coating compounds shall be resistant to fungus, weathering, ozone, high humidity, and storage at temperature up to 160 °F. The coating compounds shall be as follows (see table I).

TABLE I. Characteristics of rubber coating compounds.

Item Property	A 1/	B 2/	C 3/	Test paragraph or test method
1. Tensile strength, initial (Lbs/sq in)	1600	1500	2000	ASTM D 412
2. Ultimate elongation %	400	400	460	ASTM D 412
3. Tensile strength after immersion in distilled water @ 160 ±2 °F (% retention)	80% (min)	70% (min)	70% (min).	FED-STD-601 6121
4. Ozone resistance 20% extension 50 pphm ozone code 30 °C Time of first appearance cracking	7	7	7	ASTM D 1149

1/ Natural rubber - inside coating.
2/ Neoprene rubber- outside coating.
3/ Natural rubber reinforcing tape and sheeting.

3.4.4 Coated fabric. The elastomeric coated fabric may include a barrier to restrict diffusion. The coated fabric shall be free from blisters or pinholes and shall show no signs of coating delamination. The coated fabric shall withstand the effects of humidity, fungus, high storage temperature, ozone, and weather elements without damage, deterioration, or failure of meeting performance requirements specified herein. Unless otherwise specified (see 6.2), the exterior coating of the coated fabric shall approximate color No. 35042 Black of FED-STD-595. The elastomeric-coated fabric shall conform to the performance requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.4.5 Coatings. The material shall be coated on both sides. The exterior coating shall be neoprene. The inner coating shall be natural rubber. The coating shall meet the requirements of tables I and II where appropriate.

TABLE II. Characteristics of coated fabric.

A. Main bag (chassis) material.
B. Handle attachment reinforcement.

ITEM Property	A	B	Test paragraph or test methods
1. Weight (Oz/sq yd)	50 (min) 56 (max)	15 (min) 18 (max)	FED-STD-191 5041
2. Tearing strength warp and fill (lbs, min)	90	7	FED-STD-191 5134
3. Breaking strength warp and full (lbs, min)	360	100	FED-STD-191, 5102
4. Weathering resistance after 500 hours exposure at 5% elongation warp, and fill (percent retention of initial breaking strength (min)	80		ASTM D 822, FED-STD-191, 5804/5102 1/
5. Puncture resistance (lbs) (min)	150		4.5.2.5.1 or FED-STD-191, 5120
6. Coating adhesion (initial Natural rubber side (lbs/in) Neoprene side (lbs/in)	 25 (min) 16 (min)	 78 (min)	 4.5.2.5.2.1 4.5.2.5.2
7. Coating adhesion after immersions in distilled water at 160 ±2 °F for the following duration side natural rubber 7 days (lbs/min)	18	6	4.5.2.5.2.1 4.5.2.5.2
8. Low temperature crease resistance. (Appearance after unfolding)	No cracking or peeling		4.5.2.5.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

TABLE II. Characteristics of coated fabric. (Cont'd)

A. Main bag (chassis) material.
B. Handle attachment reinforcement.

ITEM Property	A	B	Test paragraph or test methods
9. Fungus resistance	No cracking blistering or de-lamination of coating. Retention of breaking strength 60% (min).		FED-STD-191, 5762 2/
10. Blocking	Specimen to separate within 5 seconds		4.5.2.5.5

1/ Specimens shall have exterior coating fabric carbon arc. Alternate corex D filters shall be removed.
2/ Except that the specimens shall be prepared in accordance with FED-STD-191, method 5102 and the number of specimens shall be reduced from 40 to 5 warp and 5 fill. Leaching of specimens is unnecessary. The specimens shall be exposed to the soil for eight weeks.

3.4.5.1 Coating adhesion. Samples of coated fabric shall be bonded face-to-face to provide specimens for determining adhesion between the nylon cloth and exterior coating(s); between the nylon cloth and interior coating(s); between laminations of interior coatings and barrier (if used); and between lamination of exterior coatings. In forming this bond, the specimens shall be subjected to no heat or pressure other than that normally encountered in curing the coated fabric, except for minimal pressure necessary to insure contact while the bond is setting.

3.5 Bag design.

3.5.1 Bag chassis. The main waterproof section of the bag, herein referred to as the bag chassis, shall be within the dimensions shown in table IV and figure 6. The lid shall be secured with a waterproof zipper (see figure 4) and shall have suitable molded rubber valve ports (see figure 4) for the buoyancy control equipment see 3.4.4). The bag shall be constructed of a coated fabric as specified in 3.4.4 and table II, and assembled as shown in figure 7b.

3.5.2 Exterior pockets. Nine exterior, non-waterproof pockets shall be supplied in the arrangement shown in figures 1 and 7a on the bag chassis (see table IV). The pockets shall be accessed by a top flap which shall be secured by a black acetal pinch-release buckle (figure 1 item 38) such as ITW Nexus model or BSR model 1 inch or 1-1/2 inch or equivalent. The female portion of the buckle shall be attached to the flap of the pocket with 1 inch or 1.5 inch nylon webbing measuring approximately 3 inches long. Attached to the pocket itself shall be a 1.0 or 1.5 inch wide nylon webbing measuring approximately 8 inches in length

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

which shall be folded and sewn to prevent the female portion of the fastener from slipping off of the nylon webbing that shall be attached to the pocket. This system shall be used to enable a variation in tension for the pocket flap. Provision shall be made for adequate drainage from pocket. The pocket shall be fabricated from the same material as the bag chassis. Pockets shall be designed to withstand double the design weight of the bag when used as a handle to its joining surface.

3.5.3 Frame pocket. The waterproof bag shall be equipped with a pocket, mounted so that the open end faces down to accept the pack frame, as shown in figure 3 item 6. The pocket shall be of a width and depth (normal to bag surface) to accept the frame, pack, ground troops, riveted, and shelf, cargo support (lightweight), LC-1; MIL-F-43834. The height of the frame pocket shall be 4.5 inches. The frame pocket shall have an opening, as shown in figure 3 item 6, centered on the upper seam. The opening shall have the dimensions 6.2 inches wide and extend 1.3 inches below the top seam of the flap, such that the frame of MIL-F-43834 shall be accessible through it. The top edge (measured inside the pocket) shall be 21.0 inches from the bottom of the bag. The pocket shall be designed to support three times the design weight when applied vertically or horizontally.

3.5.4 Frame attachment loops. Frame attachment loops shall be vulcanized to the bottom reinforcement on lower back parallel to the lower inner D-rings (see figure 3 item 40) centered on the lower seam with a distance of 4.3 inches apart and 19.6 inches down from the edge (inside ALCE frame pocket) to their center line. The material shall be as specified in table II, column A, with a width of 0.84 inch and have a loop of diameter 0.31 in. Each loop shall be able to support the gross weight of the bag.

3.5.5 D-rings. D-rings shall be supplied on the bag chassis in the positions shown in figures 1-3 and 7. The upper D-rings shall have their central axis on a line with that of the zipper as shown in figures 1-3. Any one of the load D-rings shall be capable of suspension supporting twice the gross weight of the bag loaded. The outer lower D-rings on the back shall have their straight axis parallel, vertical and spaced 12.0 inches apart, centered with respect to the bag (see figures 3 and 7a, item 8). The straight axis of the upper three load D-rings shall be in line with the top seam of the frame pocket, equally spaced on center in the top opening (see figure 3, item 7).

3.5.6 Zipper. The bag chassis shall be sealed with a heavy duty zipper. The zipper shall be a 30 inches long and will extend around the front and sides of the bag as shown in figures 1-5. The zipper shall have a single nonreversible bronze alloy slider. The slider shall be equipped with a stirrup pull with thong (see figure 3). The elements (teeth) shall be nickel alloy elements and riveted. The zipper tape shall be attached to the bag on the inside. Then it shall be coated with uncured rubber tape (see table III) entirely covering the edges to form a watertight seal. The zip end D-rings (see figure 3, item 13) in line with the zipper shall be positioned not more than 2.0 inches from each end of the zipper and 3 pairs of D-rings shall be attached in line with the zipper to aid opening and closing (see figure 1, item 14).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.5.7 Strap handles. There shall be three strap handles located as shown in figures 1, 2, and 3. The rear handle shall be centered above the frame pocket. The side handles shall be centered with respect to both height and depth on the sides. Each strap shall be a minimum of 1.5 inches wide and be accessible to a divers gloved hand. Each handle shall be capable of supporting three times the design weight of the bag (see 3.13).

3.5.8 Air supply system. The waterproof bag shall be supplied with an air supply system as specified herein (see figure 7a, 8 and 9). The system will be independent of the buoyancy of the users by adding air from a supply cylinder on the bag without the use of his own air supply. The system shall consist of an air cylinder and a with a 5/8 inch thread valve and connector (see figure 9), attached to a rubber gooseneck. The cylinder and all fittings shall be oxygen clean. A filling adapter (see figure 10) shall be supplied with each bag.

3.5.8.1 Cylinder. The cylinder shall be a steel or aluminum high pressure air cylinder. It shall have a maximum volume of 61.1 cubic inches of water and a minimum of 20.55 cubic inches and have a safe working pressure of 2900 psi. The cylinder's external diameter shall be a maximum 3.54 inches for 61.1 cubic inches and 2.7 inches for 30.55 cubic inches. The cylinder's external length shall be a maximum of 11.0 inches long 61.1 cubic inches and 9.84 inches for 30.55 cubic inches. The cylinder shall be DOT approved.

3.5.8.2 Cylinder valve. The cylinder valve shall be a two piece stem regulating valve made of brass with seat sealing of acetal material. The operating handwheel shall have an external diameter of 1.88 inches and provide a firm hold to a divers gloved hand. The axis of stem must be at a 30 degree angle of a 5/8 inch thread fitting axis to provide easy accessibility to the user during operations (figures 7a and 9).

3.5.8.2.1 Safety relief device. The valve shall have a safety relief device set to a maximum pressure of 3263 psi. The valve shall also have a burst disc resisting a maximum pressure of 3480 psi.

3.5.8.2.2 Safety pressure device. The valve shall be furnished with a minimum safety pressure device which will prevent the cylinder pressure from dropping below 44 psi, thus preventing entry of water into the cylinder when it is empty.

3.5.8.3 Cylinder pocket. The waterproof bag shall be equipped with a pocket to hold the air supply cylinder (see figures 3 and 7a). The pocket shall be adjustable to fit a range of cylinder sizes. The pocket shall be a sleeve type (no bottom). The pocket shall be secured in such a manner as to permit loosening and removal of the cylinder with no special tools. Laces are preferred (figure 3). Laces cord shall be laced through five pairs of eyelets (item 39). The cord shall be secured with a cord pinch release buckle. The pocket shall be hot vulcanized to the bag chassis such that it will withstand the design weight of the bag applied normally to its joined surface. The pocket shall hold the cylinder securely without rotation. The pocket shall be located as shown in figures 3 and 7a.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.5.8.4 Connector. The connector shall be a hand wheel type 5/8 inch external thread. The connector shall be made of brass with an insert socket made of acetal. A restricting check valve resisting a minimum pressure of 29 psi shall be incorporated in the connector. The valve shall prevent water entry into the bag compartment at the air gooseneck with a jubilee clip (see figure 7a).

3.5.8.5 Filling adapter. The filling adapter (figure 10) shall be made of brass. It shall be equipped with a hand wheel type 5/8 inch external thread and be O-ring sealed to the cylinder valve or scuba manifold. The filling adapter must have a pressure relief device installed to allow pressure relief for disconnection after filling the bottle.

3.5.8.6 Gooseneck. The waterproof bag will be supplied with a molded rubber gooseneck, as shown in figure 7a item 20. The fitting shall be attached to the bag to withstand bag working pressure and admit air from the supply cylinder.

3.5.9 Buoyancy adjustments. The bag shall be equipped with buoyancy control valves. The valves shall be positioned on the top flap of the bag as shown in figure 1. They shall be attached by the bag capturing lid molded rubber valve ports between the thread coupling and shall be capable of rotation in any direction. The valves shall not be cemented to the bag.

3.5.9.1 Inlet valve. The inlet valve (figures 1 and 11, item 24) shall be constructed to prevent water from entering the bag when the hose is disconnected as well as when the inlet valve is depressed while the hose is disconnected under water.

3.5.9.2 Exhaust valve. The exhaust valve shall be as specified by figure 12. The valve shall be located as shown in figures 1 and 2, item 25, and be adjustable for an opening pressure between 1.29 inches and 9.85 inches water column.

3.5.10 Inflator hose assembly. An inflator hose assembly shall be supplied with the bag. It shall be as shown in figure 13. Connector socket shall have UNF 3/8 inch 24 and have an orifice safe for breathing restricting air flow to a maximum of 2.8 cubic feet/minute at 87 psi. The hose shall be equipped with a quick disconnecting coupling as shown in figure 13.

3.5.12 Oral inflation assembly. The bag shall be supplied with an oral inflation tube positioned as shown in figure 1, item 29. The valve shall be as shown in figure 14. The tube shall be cemented to the bag lid. An inflation tube retaining strap (figure 1, item 28) shall secure tube on the bag lid.

3.5.13 Load securing/compressing straps, internal. Straps to secure and compress load shall be attached on the inside of the bag as shown in figure 5. Each strap shall loop through 4 rectangular rings (see figure 5, item 31) which shall be attached to the inside of the chassis. Each strap shall carry the design weight of the bag. An acetal strap buckle (figure 5, item 33) shall be securely sewn in one end.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.6 Construction. The SOWB shall be fabricated from single ply coated fabric as specified herein. The configuration and dimensions of the SOWB and the location of all the attachments, accessories, and appurtenances shall be as shown in table IV and figures 1 through 9. SOWB both interior and exterior, all coated fabric edges (seams, chafing patches, fabric) shall be covered to a thickness of not less than 10 mils, with coated fabric or coating compound. Coated fabric panels may be spliced together to make up the length required for a SOWB panel. All splices shall conform to all of the requirements of seams as specified in 3.6.1. There shall be no other permanent or temporary joint located over the spliced area. All such panel splices shall be located to the back side of the SOWB (see figure 3).

3.6.1 Seams. The seams between adjacent panels or fabric shall be sewn with nylon stitching and sealed with rubber tape, and they shall be so constructed as to prevent wicking of water through the nylon fabric. Parallel splices and seams shall not coincide with pockets, handles or D-rings, except if separately attached on reinforcements. Waterproof rubber or neoprene tape or material shall be applied over the seams, on both sides of the seam, so that no stitches are exposed (see figure 7). All seams shall conform to table III.

3.7 Underwater performance. The SOWB shall remain waterproof when submerged to a depth of 50 feet in sea water for a minimum period of 15 minutes (see 4.5.2.4), and when stored at a depth of 30 feet in sea water for a period of 24 hours (see 4.5.2.2). The SOWB shall be capable of floating on the surface of sea water with a load of 100 pounds (see 4.5.2.4.1). The SOWB shall be capable of neutral buoyancy at varying water depths (see 4.5.2.4.2 and 4.5.2.5.7) and shall have the capability to achieve surface floatation from a depth of 50 feet (see 4.5.2.4.3).

3.8 Maintainability.

3.8.1 Special tools. Special tools required for operational adjustments shall be placed within the equipment in a readily accessible location. The special tools will be provided with the SOWB.

3.8.2 Emergency repair. The SOWB shall exhibit no degradation in performance after maintenance operations using emergency repair parts furnished with each bag. The emergency repair kit and repair kit components shall conform to MIL-R-22368 and be provided by the manufacturer with each bag.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

TABLE III. Characteristics of seams.

Item Property	Requirement	Test Method
1.Breaking strength (initial) (lbs/in)	240 (min)	FED-STD-601,8317 or ASTM D 751
2.Breaking strength after immersion is distilled water @ 160 ±2 °F for the following duration		
7 days	200 (min)	FED-STD-601, 8311 6001
14 days	175 (min)	FED-STD-601, 8311 6001
3.Dead load shear resistance under 50lb/in stress @ 180 °F for 8 hours	0.1 inch slippage (max)	4.5.2.5.6
4.External peel adhesion (initial) (lbs/in)	10 (min)	ASTM D 413 machine method
5.External peel adhesion after immersion in distilled water @ 160 ±2 °F for the following durations:		
7 days (lbs/in)	8 (min)	ASTM D 413 machine method
14 days (lbs/in)	8 (min)	

3.9 Performance characteristics. The SOWB and components shall withstand folded storage at temperatures from 160 to -30 °F without damage. The SOWB and components shall be suitable for operational use at temperatures from +125 to -25 °F. The bag shall not be damaged when exposed to fungus growth or relative humidity up to 100 percent such as is encountered in tropical climates. The SOWB shall be suitable for use in continuous contact with rainwater or ground water. There shall be no evidence of leakage or seepage when the SOWB is filled with air and completely immersed at 50 feet in sea water (FSW). The SOWB shall be capable of withstanding an internal pressure of 10 pounds per square inch (psi), without evidence of leakage. The filling adapter shall withstand a hydrostatic pressure not less than 3480 psi without leaking.

3.10 Human factors. The bag shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special emphasis shall be given, but not limited to, general requirements (4), valve controls (5.1.2.4), labeling (5.5), configuration (5.11.1.1.3), transportability by personnel (5.11.1.2), edge rounding(5.1.2.3 thru 5.1.2.5, 5.13.5.4), and equipment thermal hazards (5.13.4.6) as applicable to freezing caused by rapidly expanding air.

3.11 Environmental requirements. The SOWB shall conform to temperature limits as described in 3.9 and tests as described in 4.5.2.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.12 Dimensions. The dimensions of the SOWB shall be as shown in table IV.

TABLE IV. Dimensions.

Item	Inside Dimensions
Main Bag (top)	H=24 in X L=15.7 in X W=8.66 in Radius = 4.33 in
Main Bag (bottom)	H=24 in X L=15.9 in X W=9.25 in
Pockets (figure 7, item 4)	H=7.9 in X D=2.0 in X W=3.8 in
Pockets (figure 7, item 5)	H=9.5 in X D=2.10 in X W=4.01 in

3.13 Weight ratings. The weight of the bag shall not exceed 15 pounds excluding bottle, control valve, and gooseneck adapter. The gross weight rating for each loaded bag shall be 80 pounds, not including the frame. The design weight of the bag as referred to herein is 80 pounds which is the gross weight of the bag fully loaded.

3.13.1 Tare weight. The tare weight shall be the average of the first 20 production units. This weight shall be marked on the bag (see 3.15). The weight of the remaining production units shall be within 0.5 pound of the marked tare weight.

3.14 Finish. Unless otherwise specified (see 6.2) the SOWB finish coating shall conform to 3.4.3.

3.15 Identification and marking. The SOWB shall be identified in accordance with MIL-STD-130 by means of an identification label. The bag shall have the following information on a separate rubber tag in order as follows:

Design MIL-B-53083, FSCM 97403
Manufacturer , FSCM Number
Contract No:
Waterproof Bag, Load Carrying Serial no.
Tare Weight
30 ft. depth-uses ALCE Frame Gross > or = 80 pounds
Cylinder 2900 psi/200 bar working pressure.
NSN 4220-00-XXXXXXX
Property of U.S.

The tag shall be hot vulcanized to the inside of the top lid of the bag (see figure 4). The identification marking shall be located externally on the (SOWB) on the flattest, most uncluttered surface available in such position that it is not obscured by adjacent items or subject to handling damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.16 Workmanship. Workmanship shall permit no defects adversely affecting the strength or serviceability of the finished SOWB. Any fabric-reinforced shall contain no gum voids, cracks, or tears that could adversely affect the strength of the assembly. All metal parts shall be clean and free of foreign materials. Any necessary repair or rework shall restore the reworked area to its full strength and shall meet all applicable requirements of this specification. The cemented surfaces of all spliced areas, and patch-type repairs shall effect a bond that will result in strength of the cemented area not less than the strength of adjacent bag fabric.

3.16.1 Repair or rework. Any repair or rework of SOWB shall be accomplished before the first inspection (quality conformance inspection or inspection of packaging, as applicable). Rework or repair of any seam shall not exceed 10 percent of the length of that seam. Repair or rework of panels, other than seams, shall be limited not greater than 28 square inches with one dimension not to exceed six inches. All repair and rework procedures shall be in accordance with the repair and rework plan.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection/test. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
c. Inspection of packaging (see 4.6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. The presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 level II and shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.4.3 Tests. Samples shall be selected in accordance with 4.4.1 level 5-3 and shall be tested as specified in 4.5.2. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The SOWB assembly, component, and parts, shall be examined for the following defects.

Number		Requirement Paragraph
Major		
101.	SOWB any component or part missing.	3.1
102.	Material not as specified.	3.4
103.	Materials not resistant to corrosion and deterioration or treated to be made resistant or corrosion and deterioration.	3.4.1
104.	Dissimilar metals not in accordance with MIL-STD-889.	3.4.1.1
105.	Material and material finish or treatment information not available.	3.4.1.2
106.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the SOWB.	3.4.1.3
107.	Nylon cloth not as specified.	3.4.2
108.	Coated fabric not as specified.	3.4.4
109.	Coating material not as specified.	3.4.5
110.	Bag chassis not as specified.	3.5.1
111.	Exterior pockets not as specified.	3.5.2
112.	Frame pocket not as specified.	3.5.3
113.	Frame attachment loops not as specified.	3.5.4
114.	D-rings not as specified.	3.5.5
115.	Zipper not as specified.	3.5.6
116.	Air supply system not as specified.	3.5.8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

117.	Air cylinder not as specified.	3.5.8.1
118.	Air cylinder valve not as specified.	3.5.8.2
119.	Safety relief device not as specified.	3.5.8.2.1
120.	Safety pressure device not as specified.	3.5.8.2.2
121.	Air connector not as specified.	3.5.8.4
122.	Air filling adaptor not as specified.	3.5.8.5
123.	Gooseneck molded rubber not as specified.	3.5.8.6
124.	Buoyancy control valve not as specified.	3.5.9
125.	Inflator hose assembly not as specified.	3.5.10
126.	Oral inflation assembly not as specified.	3.5.12
127.	Construction not as specified.	3.6
128.	Seam between adjacent panel or fabric not as specified.	3.6.1
129.	Special tools not as specified.	3.8.1
130.	Emergency repair kit not as specified.	3.8.2
131.	Performance requirements not as specified.	3.9
132.	Human factors not as specified.	3.10
133.	Environmental temperature requirements not as specified.	3.11

Minor

201.	Material documentation not as specified.	3.4.1.4
202.	Strap handle location not as specified.	3.5.7
203.	Air cylinder pocket not as specified.	3.5.8.3
204.	Load securing/compression straps internal not as specified.	3.5.13
205.	Dimensions not as specified.	3.12
206.	Weight ratings not as specified.	3.13
207.	Finish not as specified.	3.14
208.	Identification and marking not a specified.	3.15
209.	Workmanship not as specified.	3.16

4.5.2 Tests.

4.5.2.1 Conditions of test. Prior to beginning the tests, the contractor shall assemble the SOWB and keep components and parts ready for testing.

4.5.2.2 Test schedule.

4.5.2.2.1 Underwater integrity. The bag shall be weighted down and immersed in water at a temperature of approximately 40 ^{o}F at a depth of 30 feet sea water for 24 hours. Rotate the bag to check each surface for leakage. After removing the bag from the water inspect the inside for moisture. Evidence of moisture inside the bag shall constitute failure of this test.

4.5.2.2.2 Main compartment. Insert, without crushing, a wooden frame, box, or other suitable rectangular structure measuring 22.6 inches x 7.8 inches x 7.8 inches into the main compartment and close the zipper completely. Failure to close the zipper or to complete the insertion will constitute failure of the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.5.2.2.3 Exterior pockets. Insert a wooden block measuring 7.6 inches x 1.88 inches x 3.5 inches (pocket 5A) or 9.25 inches x 1.97 inches x 3.8 inches (pocket 5B) in each exterior pocket and buckle the pocket. Any failure to close the buckle or fully insert the block shall constitute failure of this test.

4.5.2.2.4 Frame pocket. Insert a frame conforming to MIL-F-43834 or a suitable tube of equal dimensions into the frame pocket and verify fit. Failure to fit or expose the frame through the opening shall constitute failure of this test.

4.5.2.3 Lifting tests. Each sample bag shall be loaded, lifted and suspended as specified in 4.5.2.3.1 thru 4.5.2.3.4. Any stretching or separation of fabric, plies, or attachment shall constitute failure of this test. The loading shall be by weights placed in the main compartment, and the zipper shall be closed.

4.5.2.3.1 Pocket strength. The bag shall be loaded to its design weight. The bag shall be lifted clear and suspended for not less than 3 minutes using each exterior pocket as a lifting point. Evidence of tearing or permanent deformation of any pocket or main bag material shall constitute failure of this test.

4.5.2.3.2 Tie down/lifting rings. The bag shall be loaded to its design weight. The bag shall be lifted clear and suspended for not less than 3 minutes by each D-ring except the zipper support D-rings. Evidence of tearing or permanent deformation of bag fabric or distortion of D-ring material shall constitute failure of this test.

4.5.2.3.3 Handles. The bag shall be loaded to three times its design weight (see 3.5.7). The bag shall be lifted clear and suspended for not less than 5 minutes by each handle. The lifting attachment shall contact the handle for no more than 3 inches (76.2 mm) along the handle length. Evidence of any tearing or permanent deformation of handle or bag material shall constitute failure of this test.

4.5.2.3.4 Frame pocket. The bag shall be loaded to three times its design weight (see 3.13). The bag shall be lifted clear and suspended vertically on a frame conforming to MIL-F-43834 or a tube or equal dimensions for not less than 20 minutes. Any permanent deformation of the frame or tearing or deformation of any part of the bag shall constitute failure of this test.

4.5.2.4 Buoyancy test. Any visible leaking, tears, or inability to perform the stated actions as required in 4.5.2.4.1 thru 4.5.2.4.3 shall constitute failure of these tests.

4.5.2.4.1 Float test. The bag shall be assembled with the air supply cylinder pressurized to 2900 psi. Each bag shall be loaded to 100 pounds and sealed. The bags shall be floated on water using air from the cylinder as necessary. The bags shall be stabilized and float for at least 15 minutes with no adjustments. Any visible loss of buoyancy, air leakage, or fabric damage shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.5.2.4.2 Submerged buoyancy. From the floating position, submerge the bags by removing air through the outlet valve and adding weights to the outside pockets as necessary. The bags shall be sunk to a depth of 10 feet and stabilized there, in a horizontal (swimming) position, at neutral buoyancy. They shall be left there and exhibit less than ±1 foot change in stability for 15 minutes. The same test shall be repeated at 30 feet and 50 feet depths.

4.5.2.4.3 Operational test. From the neutral position at 50 feet, connect an air supply hose to the inlet valve, inflate the bag to positive buoyancy, and allow the bag to ascend to the surface freely and unsupported. Upon reaching the surface, the bag shall be removed from the water, dried, and the inside examined. Any evidence of leakage of water shall constitute failure of the test. If leakage is due to the inlet or outlet valves, the contractor may repeat the test with replacement valves on the same bag.

4.5.2.5 Material tests. The contractor shall submit test reports or perform tests as required in 3.4.3 (table I), 3.4.4 and 3.4.5 (table II), and 3.6.1 (table III) with the following notes.

4.5.2.5.1 Puncture resistance. FED-STD-191, method 5120 applies except that the ring clamp mechanism shall have an internal diameter of 3.00 inches and the ball shall be replaced by a piercing instrument shaped like a flared, flat-tip screwdriver, having a width of .312 ±0.010 inches and thickness of 0.031 ±0.004 inches at the extreme tip. The piercing tip edges shall be rounded to a .010 inch radius. The piercing instrument shall be oriented to intercept the warp and fill threads at an angle of approximately 45 degrees. The average of three test specimens shall be reported. Nonconformance to 3.4.5 (table II) shall constitute failure of this test.

4.5.2.5.2 Coating adhesion test procedure. The adhesion shall be determined in accordance with ASTM D 413, machine method except that the specimens shall be 2 inches (50.8 mm) wide. The specimens shall be of sufficient length to conduct adhesion tests for both the initial values and after fuel or water immersions. The adhesion results obtained on each immersed specimen shall be compared with the initial adhesion of the same specimen to determine percentage of adhesion retained. The reported adhesion and percent retention shall be the average of not less than three specimens. Attempts shall be made to cut the coating back to the nylon cloth and to determine the adhesion value at the coating-to-cloth interface. However, if a specimen separates at a plane other than the bond of the coating to cloth (such as between layers of coating materials or between barrier film and coating) the adhesion value and the plane of failure shall be recorded. Immersed specimens shall be conditioned in the test fluid at 73 ±5 °F for 30 to 90 minutes before testing. Testing of immersed specimens shall be completed within 3 minutes after removal from the conditioning fluid. Immersion of specimens shall be in accordance with FED-STD-601, method 6001. Nonconformance to 3.4.4 and table II shall constitute failure of this test. Any obvious bond failure evident after immersion but before stressing, even if the plane of failure is not sandwiched between the layers of nylon fabric, shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.5.2.5.3 Seam tests. The bonding together of any two or more pieces of coated fabric (such as lap joints, butt joints, closures, chafing patches, coated fabric flanges of fittings, etc.) shall be considered as seams and shall be subjected to all seam tests specified herein, except chafing patches which will only be tested for peel. The average breaking strength of five specimens for each type seam for each test shall be reported for conformance to table III. Breaking strength specimens shall be 2 inches wide (50.8 mm) (parallel to the seam) and shall extend (perpendicular to the seam) 3 inches (76.2 mm) beyond both edges of the seam. No part of the test specimens shall be coated or covered during the fuel or water immersion periods. Specimens shall be cooled in the immersion fluids at 73 ±5 °F for 30 to 90 minutes before testing. Testing of immersed specimens shall be completed within 3 minutes after removal from the immersion fluids. The average peel adhesion strength of three specimens for each type seam shall be reported for conformance to table III. Peel adhesion specimens shall be of sufficient length to determine both the initial and after fuel or water adhesion values on the same specimen. If seam construction involves the use of binding thread, then the peel specimens shall be prepared with threads removed. Nonconformance to 3.6.1 and table III shall constitute failure of this test.

4.5.2.5.4 Low temperature crease resistance. Fold two specimens, each 8 inches square in half in each direction so that a folded corner occurs in the center of each specimen. Place each folded specimen under a 4-pound load and condition at -25 °F for 48 hours. At the end of the conditioning period, unfold the specimens while still at a temperature of -25 °F and examine visually. Signs of cracking, peeling, or delamination of any coating material shall constitute failure of this test.

4.5.2.5.5 Blocking. Place two coated fabric specimens 6 inches by 1 inch in an oven on a smooth surface in such a manner that the ends are overlapped 1 inch. Place a 4-pound weight directly on the overlapped area. After conditioning at a temperature of 158 ±2 °F for 4 hours, remove the weight and take the specimens from the oven and condition for 1 hour at 73 ±3 °F. Attach one end of the specimen in a suitable clamping device allowing the free end to hang down. Suspend a 4-ounce weight from the free end of the specimens. Inability of the strips to separate within 5 seconds under the 4-ounce load shall constitute failure of this test.

4.5.2.5.6 Dead load shear resistance. The test specimens shall be 1.0 inch ±0.020 inch wide (parallel to the seam), and coated fabric shall extend a minimum of 3 inches (perpendicular to the seam) on each side of the seam. One index mark shall be scribed on each side of the seam to facilitate observation and measurement of slippage. Each specimen shall be subjected to a constant (dead load) tension force of 50 pounds ±1/2 pound at 180 ±5 °F. After 8 hours examine each specimen while still under tension for signs of slippage or separation. Three specimens shall be tested for each determination. Slippage, by any specimen, greater than specified in 3.6.1 and table III shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.5.2.5.7 Underwater storage test. The bag shall be loaded with its normal weight breakdown (40 pounds dry, 20 pounds wet). It will then be taken to a depth of 30 feet seawater (or equivalent fresh water) where it will be made naturally buoyant by adding air from the supply cylinder and bleeding through exhaust valve as necessary. Once natural buoyancy is achieved the bag will remain at 30 feet for 24 hours with no more than a ±2 feet change in depth within the 24 hour period. At the end of 24 hours the bag will be brought to the surface, dried on the outside and opened. The interior will be checked for any sign of leakage. Failure to stabilize, to maintain depth or to maintain watertight integrity shall constitute failure of this test.

4.5.2.5.8 Low temperature environment. The bag shall be folded or rolled to a size suitable for placing in the shipping container and then placed in a low temperature environment of -30 ±2 °F, for a period of 24 hours. The environmental temperature shall then be increased to -25 ±2 °F, for an additional 24 hours. At the end of this period, while still at -25 °F, the bag shall be slowly unfolded in not less than 5 minutes nor more than 15 minutes. Flaking, cracking, or separation of the coated fabric shall constitute failure of this test.

4.5.2.5.9 High temperature environment. The procedure for the high temperature test shall be the same as that specified in 4.5.2.5.8, except that the temperature shall be at 160 ±2 °F for a period of 24 hours. Damage to the coating or separation of surfaces shall constitute failure of this test.

4.5.2.5.10 Nylon cloth. Nylon cloth properties shall be tested in accordance with FED-STD-191, method 5804 and 5104. Properties shall apply after heat setting. In method 5804 alternate corex D filter shall be removed. Specimens shall be raveled for method 5104 after 100 hours of accelerated weathering. The initial breaking strength warp and fill (lbs. [min.]) shall be determined. The edges of the test specimen shall be coated by dipping or brushing with a nylon solution or other adhesive or sealed by melting with heat sufficiently to preclude yarn slipping while under test. Breaking strength after accelerated weathering less than 45 percent of the initial breaking strength shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

134	Preservation not as specified for level A or C (see 5.1).
135.	Shipping containers not as specified for level A, B or C (see 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

136. Strapping not as specified for level A or B (see 5.2).
137. Marking incorrect, illegible or missing (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. The empty stainless steel cylinder shall be placed and secured in its pocket. Technical manual shall be preserved in accordance with method IC-1 or IC-3 or MIL-P-116, and placed inside the bag. The bag zipper shall be closed. The bag shall be put in a box in accordance with PPP-B-636, type CF, class weather resistance, variety single wall, grade V3c.

5.1.2 Level C. The instruction manual shall be placed inside the bag. The bag zipper shall be closed. The bag shall be put in a box in accordance with PPP-B-636, type CF, class domestic, variety single wall, grade 275.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. A quantity of bags, preserved as specified in 5.1, shall be packed in a container in accordance with PPP-B-601, overseas type, style optional. The maximum gross weight shall not exceed 250 pounds. The boxes shall be strapped with steel strapping in accordance with QQ-S-781, finish B.

5.2.2 Level B. The base shall be packed as for level A; except the box shall be domestic type and the strapping shall be finish A. As an alternative, a quantity of bags shall be packed in a container in accordance with PPP-B-640, class weather resistant, style optional.

5.2.3 Level C. The bags preserved as specified in 5.1 shall be consolidated in a fiberboard shipping container in a quantity not to exceed the weight limitation of the container. As an alternative, the preserved bags may be palletized.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. SOWBs are intended to carry 60 pounds of equipment, 40 dry, and 20 wet, for underwater transportation to a depth of 50 FSW having air supplied during the descent. Small non-waterproof pockets are added on the SOWB to carry small items not affected by soaking, and to assist in submerging.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DODISS applicable and exceptions thereto (see 2.1.1).
c. When a first article is required, time frame for submission and number of units for testing.
d. When additional first article samples are required (see 3.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

e. Time frame when process data is to be made available (see 3.3).
f. Color when other than as specified (see 3.4.4).
g. Finish when other than as specified (see 3.14).
h. Level of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, tests and approval of the first article test results and disposition of the first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, including installation procedures, instructional materials, illustrated parts lists, early delivery of SOWB systems for training development, and contractor's maintenance and operational manual to be furnished with each assembly.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of instruments.

6.6 Definitions. The following definitions apply to the text of this specification.

6.6.1 Blister. A blister is a void or hole, which causes protrusion on a surface when hot, may not show on a surface when cold, and may be covered or open.

6.6.2 Pinhole. A pinhole shall be defined as minute circular void or solvent blow hole.

6.6.3 Rework. Rework shall be defined as an operation performed during a production operation due to an inadequacy or error during that operation.

6.6.4 Repair. Repair shall be defined as a corrective operation which is required because of an omission of a step or incorrect performance of a step or process during a previously completed manufacturing operation.

6.7 Subject term (key word) listing.

Air cylinder
Buoyancy adjustment
Elastomeric coated fabric
Load securing straps
Oral inflation assembly
Seams
Special operations waterproof bag (SOWB)

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 4220-A311

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

REFERENCE LIST

ITEM	DESCRIPTION	#	FIG(S).
1	Chassis body	1	7a, 7b
2	Chassis lid	1	7a, 7b
3	Chassis bottom	1	7a, 7b, 4
4	Exterior pockets	6	1, 7a, 7b
5	Exterior pockets	3	1, 7a, 7b
6	Frame pocket	1	3
7	Load D-ring back	3	3
8	Lower outer back	2	1, 2, 3, 7a
9	Lower inner back	2	1, 2, 3, 7a
10	Lower outer front	2	1, 2, 3, 7a
11	Zipper	1	1, 5, 7b
12	Stirrup pull thong	1	1, 2, 3
13	Zipper end D-ring	2	3
14	Zipper aid D-ring	6	1
15	Strap handles	3	1, 2, 3
16	Cylinder	1	8
17	Cylinder valve	1	7a, 8
18	Connector	1	7a, 9
19	Jubilee clip	1	7a
20	Gooseneck	1	7a
21	Cylinder pocket	1	3, 7a
22	Lace cord	1	3, 7a
23	Lace cord buckle	1	7a
24	Inlet valve	1	1, 11
25	Outlet valve	1	1, 7b, 12
26	Inflator hose	1	13
27	Oral inflator valve	1	1, 7b, 14
28	Retaining strap	1	1
29	Oral inflator tube	1	1, 7b
30	Load securing straps	2	5
31	Rectangular rings	8	5
32	Rectangular ring	8	4, 5
33	Acetal buckle	2	5
34	Valve ports	2	4
35	Rubber tape		7b
36	Rubber foil	2	7b
37	Lid reinforcement	1	7b
38	Pinch-release buckle	9	1
39	Lace eyelets	10	3
40	ALCE frame loop	2	3
41	ID label	1	3, 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

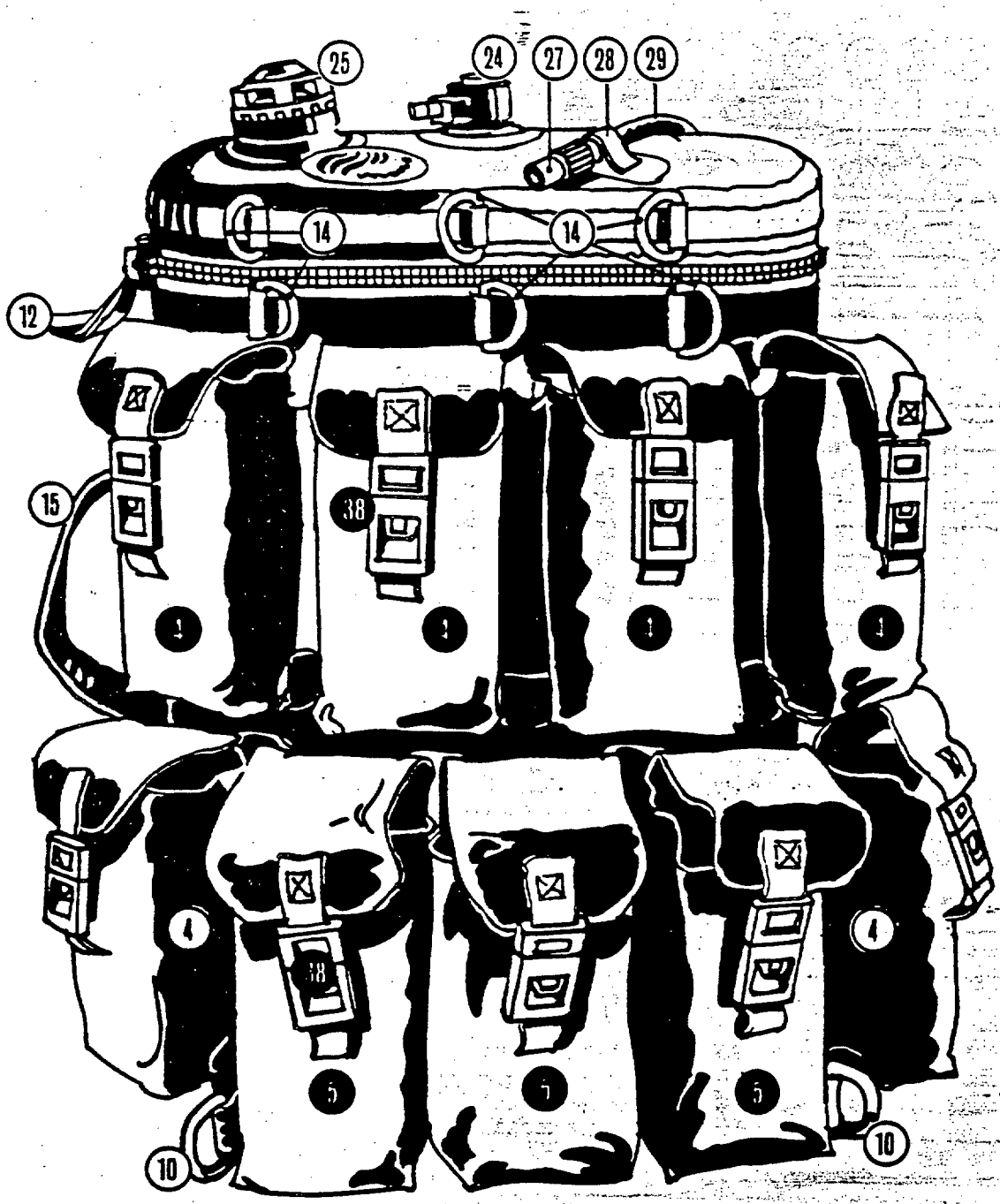


Fig 1 front

X-4805

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

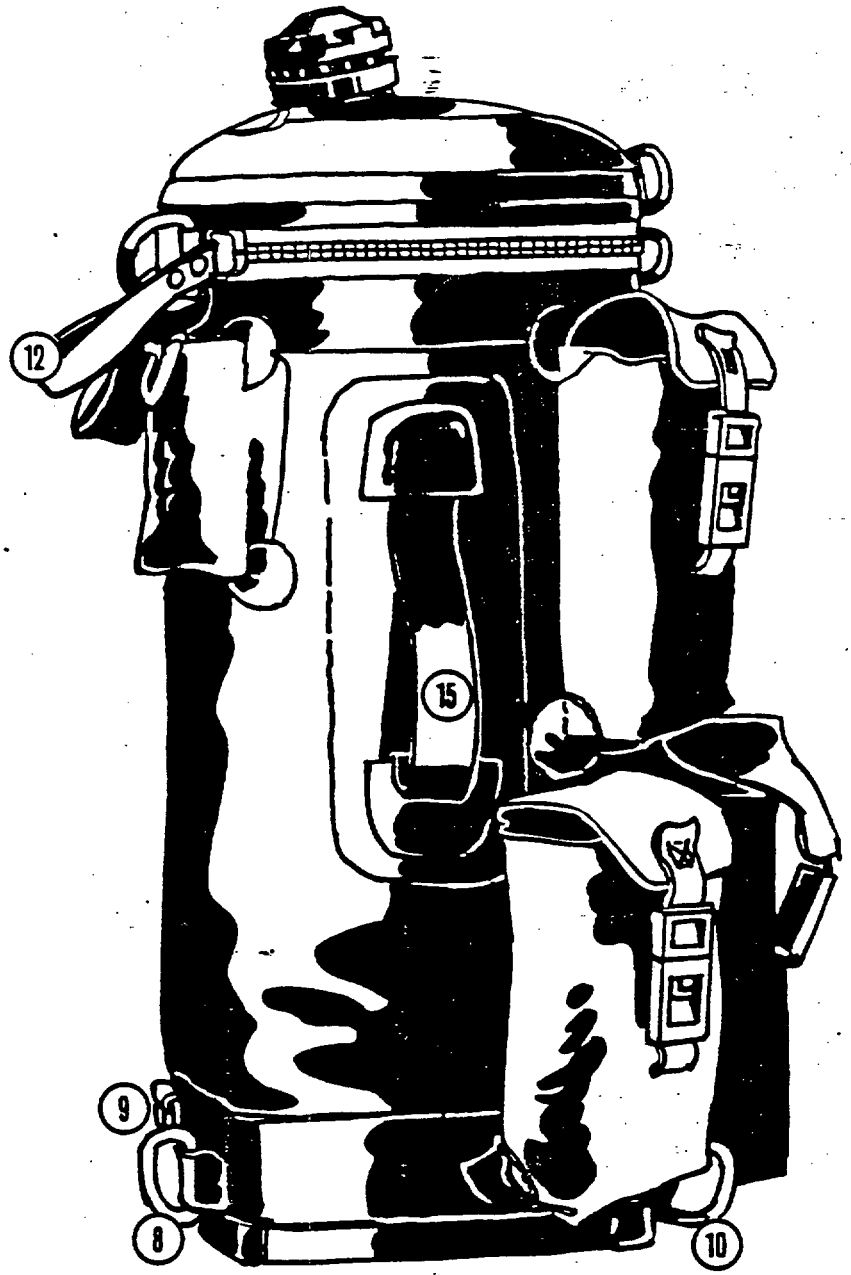


Fig 2 left

X-4806

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

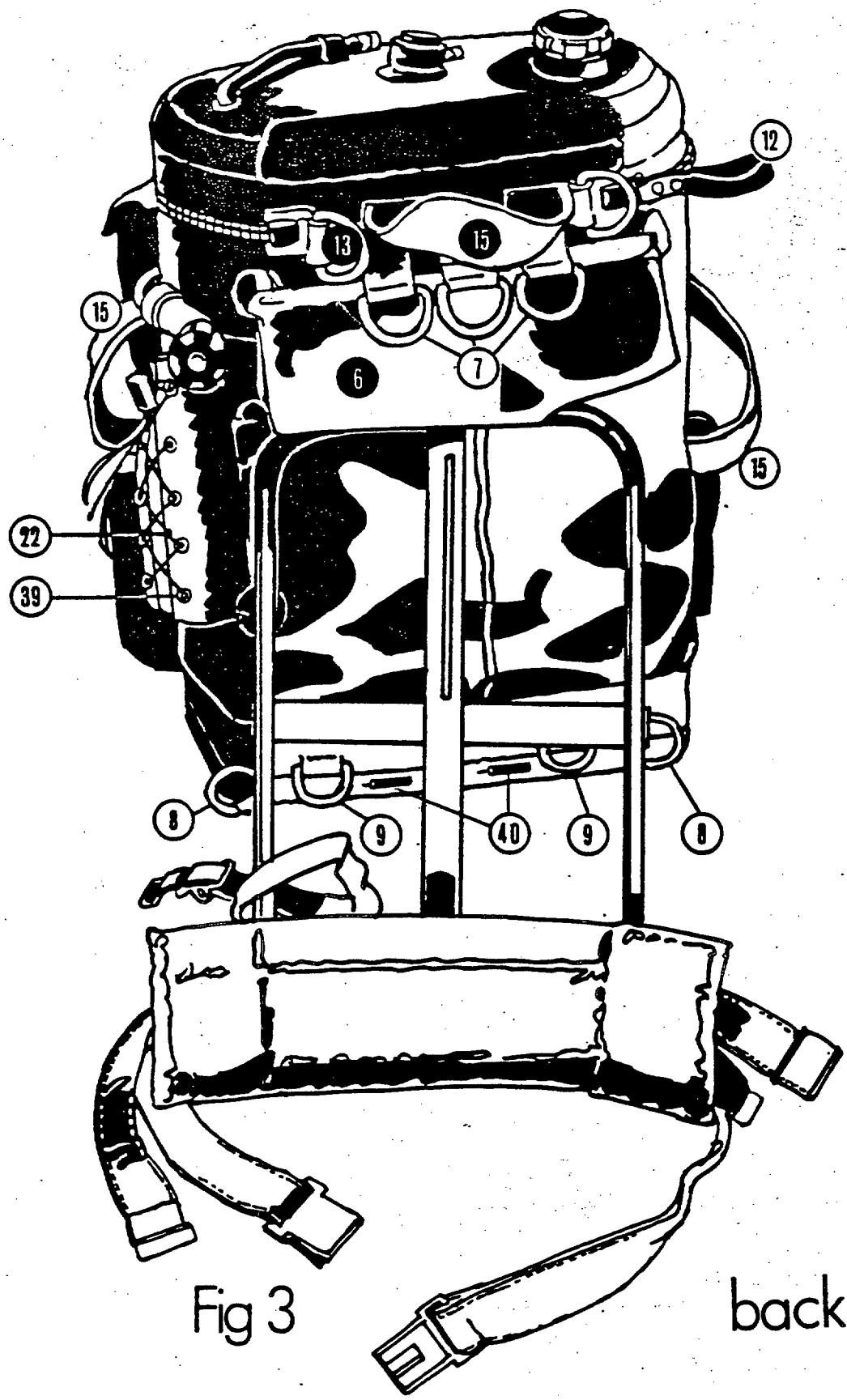
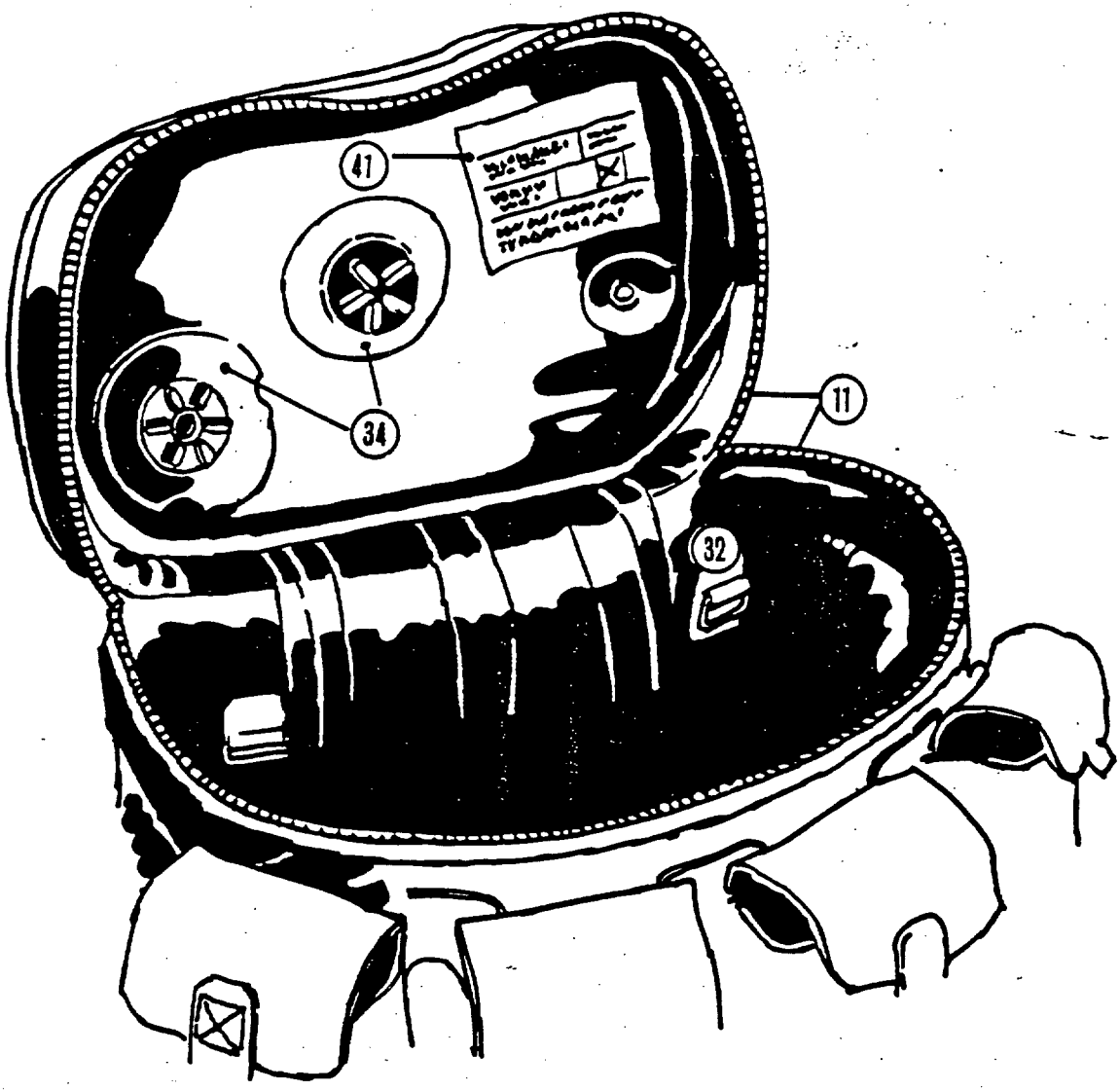


Fig 3

back

X-4807

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.



Open lid inside

Fig 4

X-4808

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

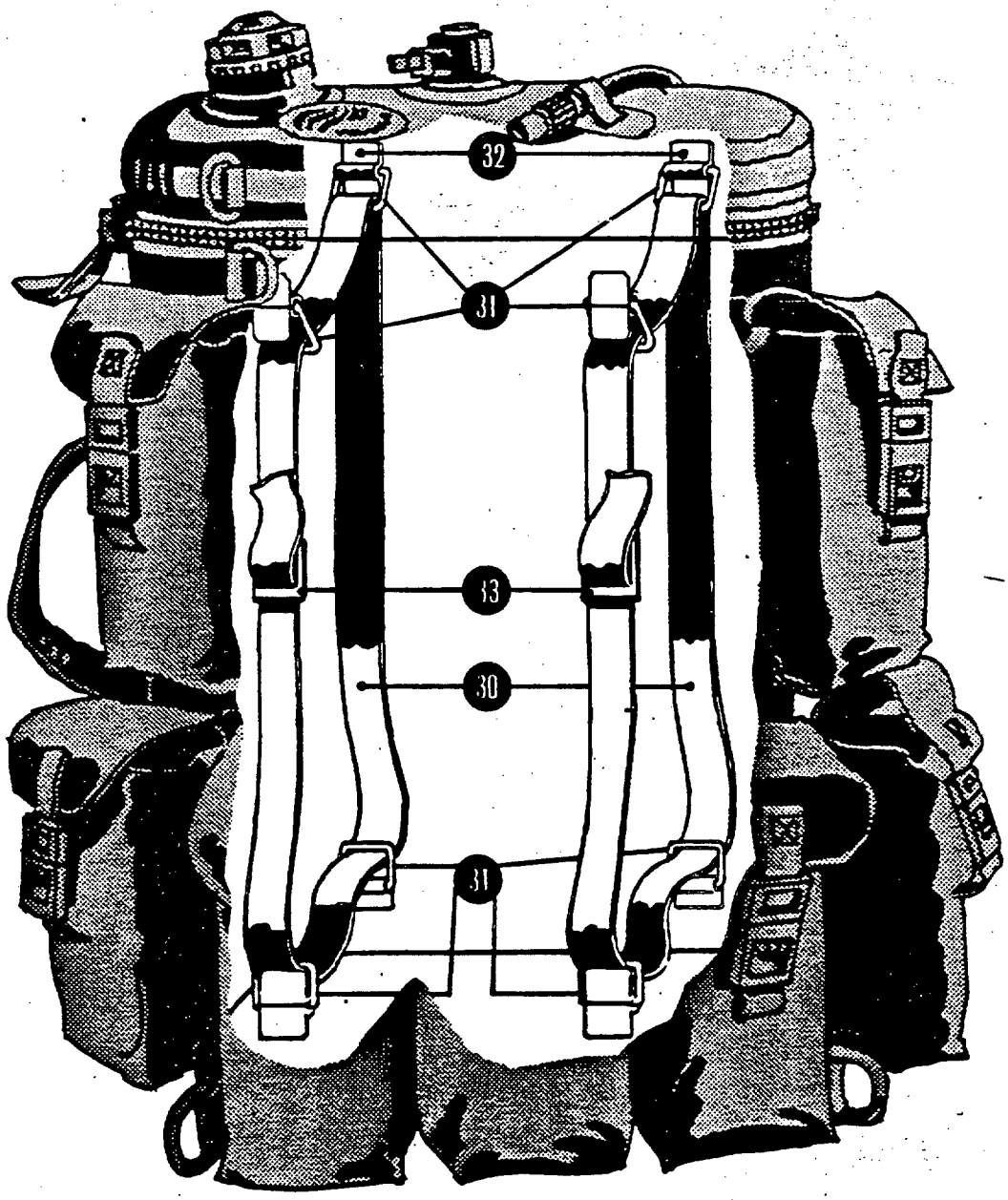
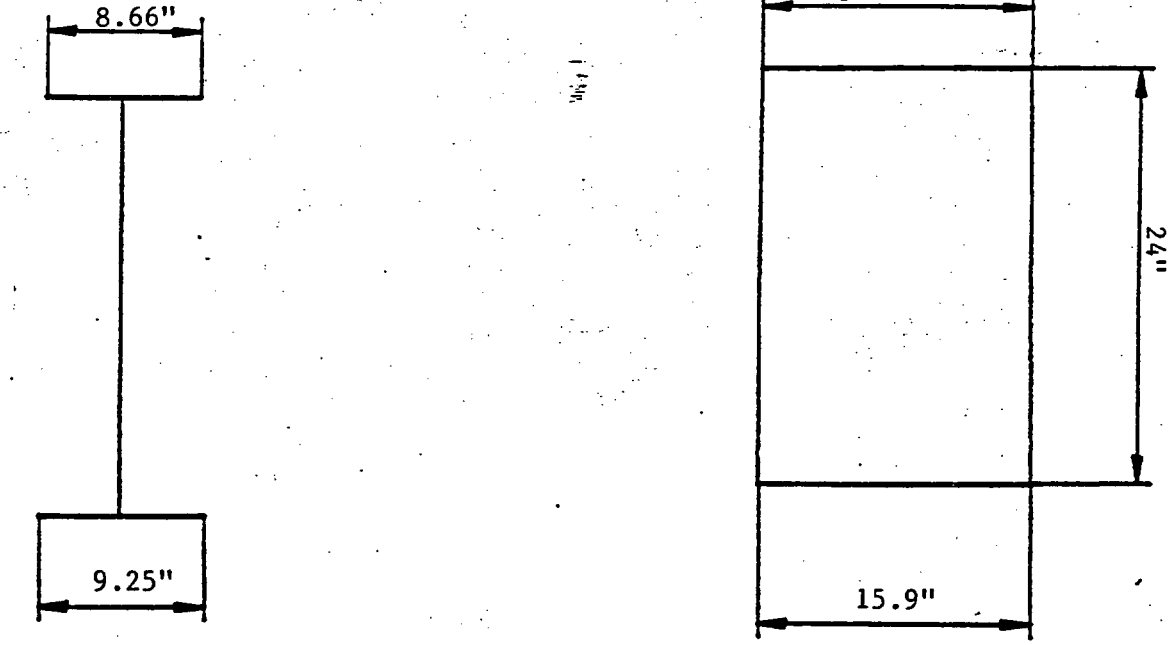


Fig 5

X-4809

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.



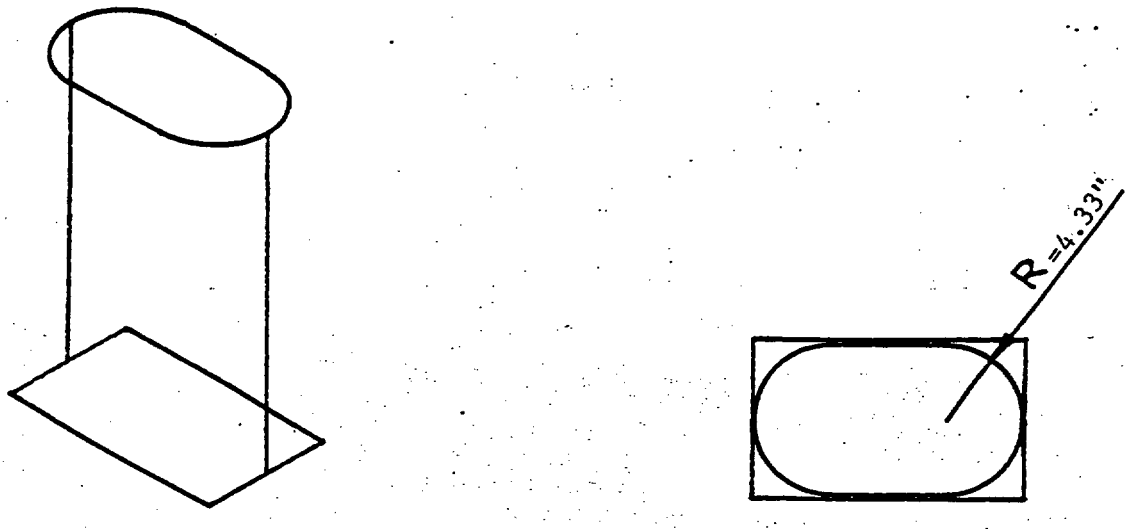


fig.6

scale.1:10	CHASSIS
	X-4810
SPECIAL OPERATIONS WATERPROOF BAG	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

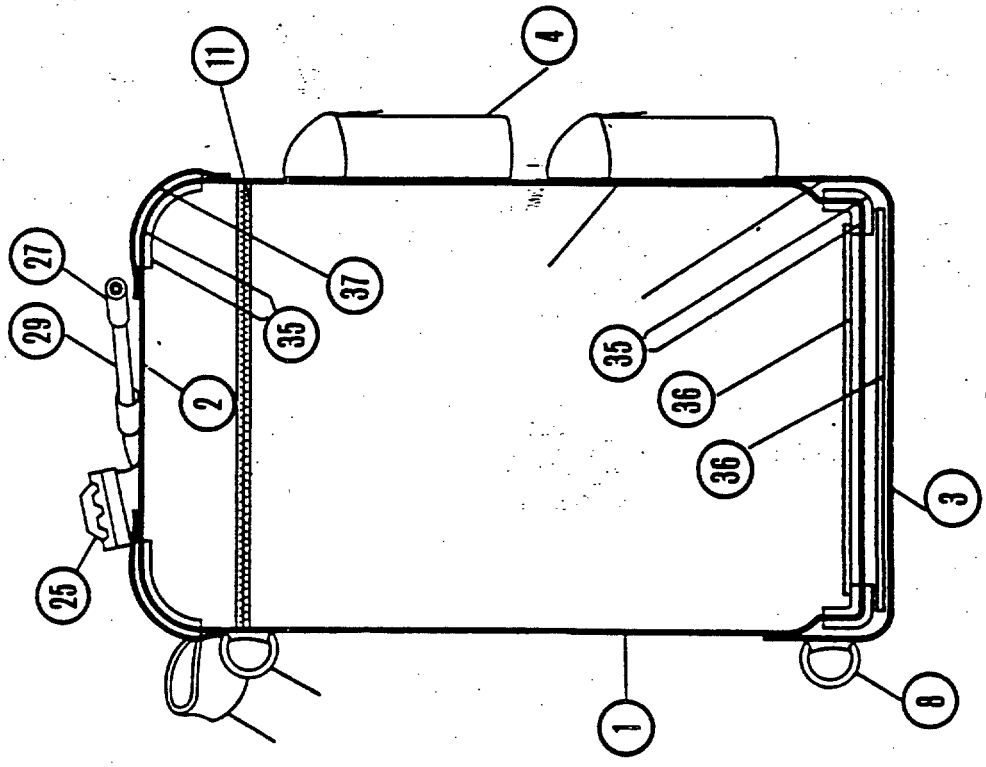


Fig 7B left

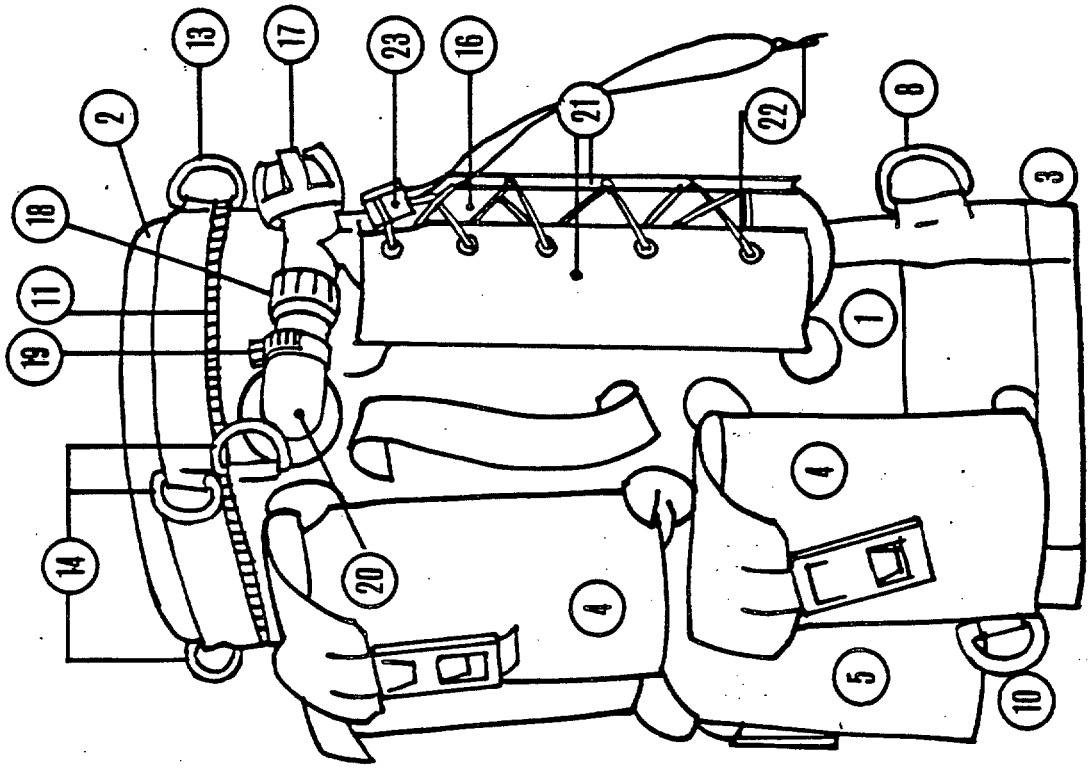


Fig 7A back-right

X-4811

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

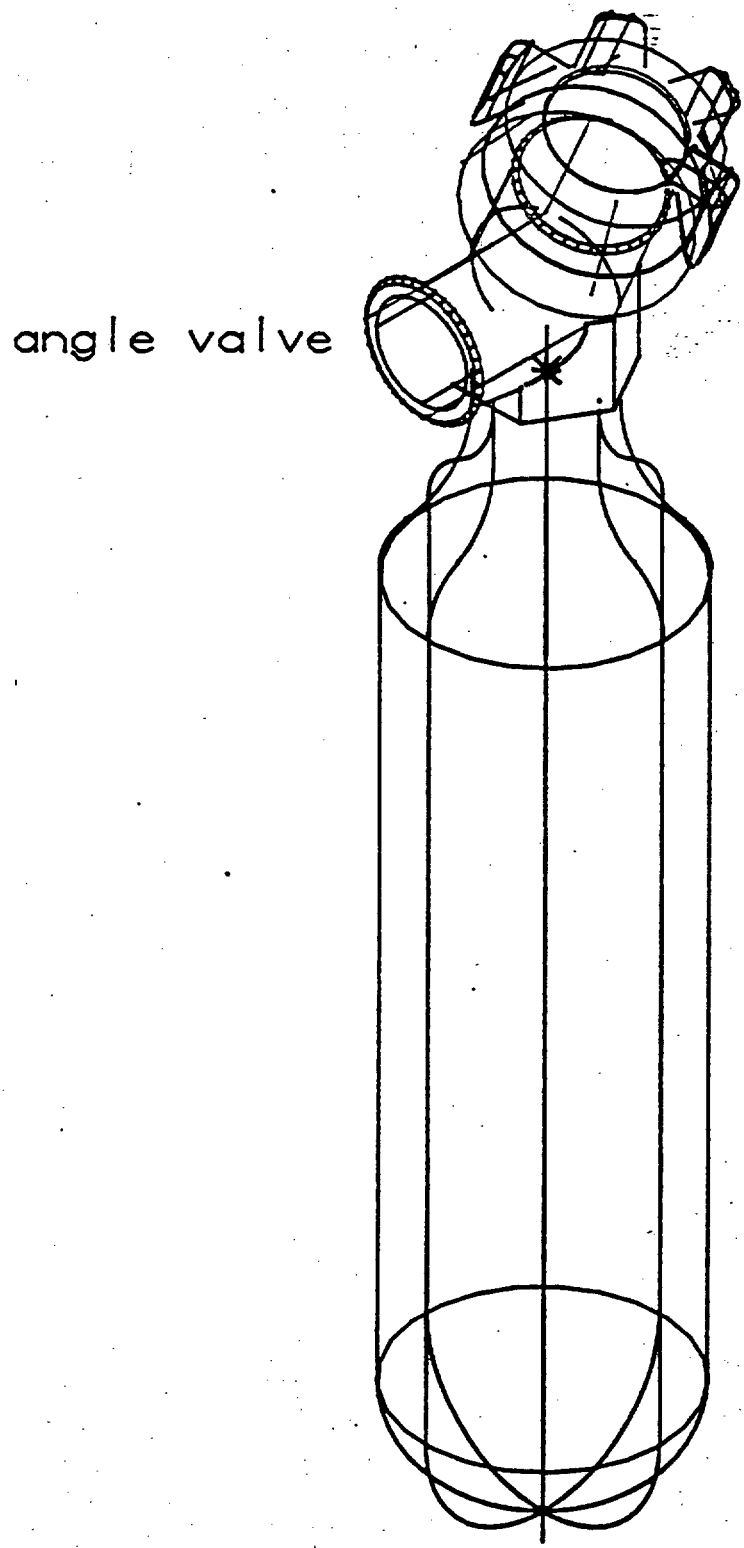


fig.8

ANGLE CYLINDER VALVE

SPECIAL OPERATIONS WATERPROOF BAG

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

X-4812

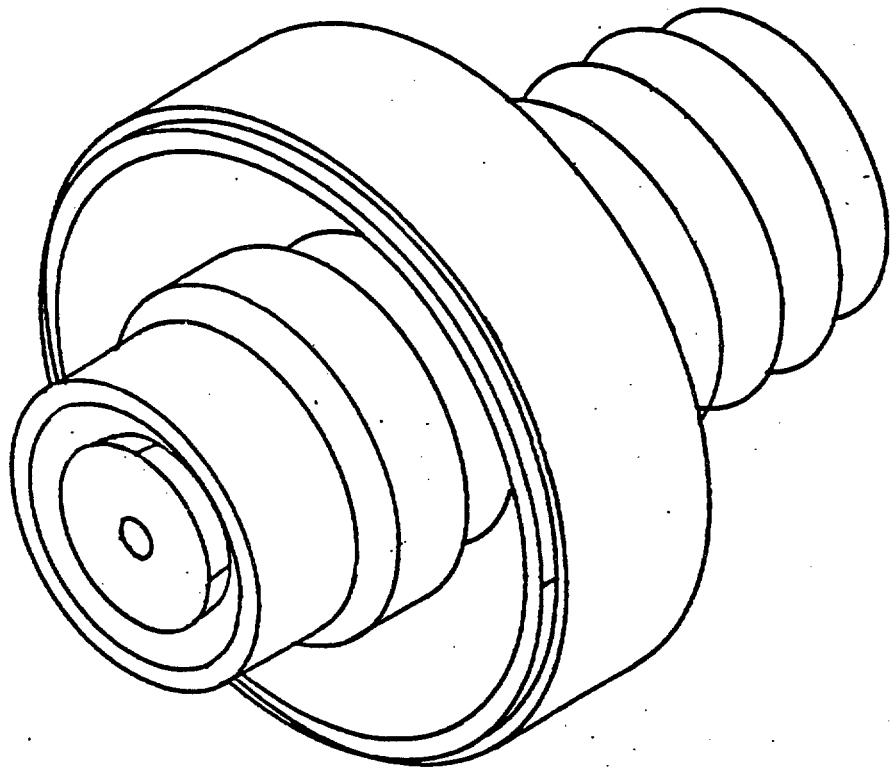


fig. 9

CONNECTOR

X-4813

SPECIAL OPERATIONS WATERPROOF BAG

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

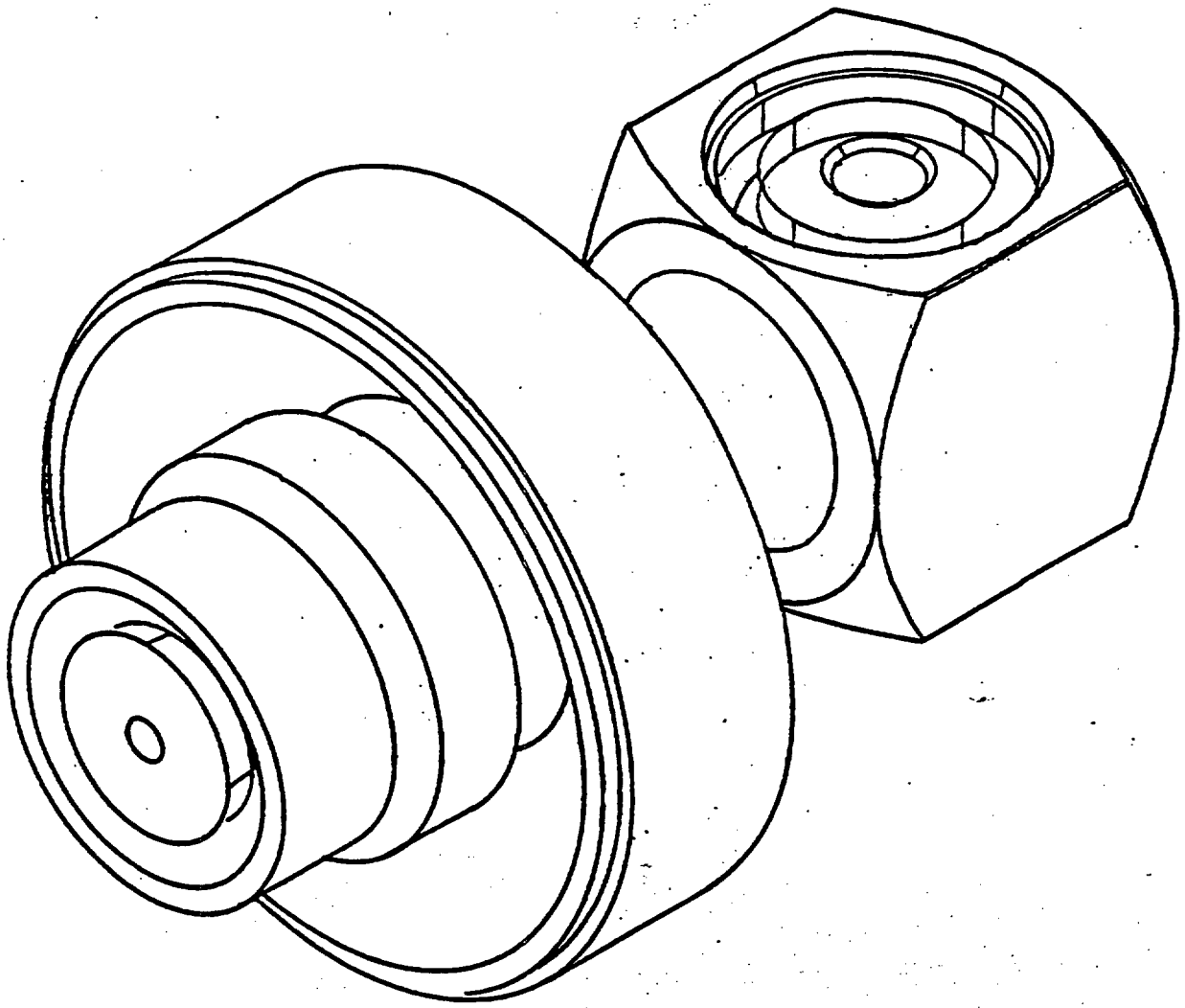


fig.10

FILLING ADAPTER

X-4814

SPECIAL OPERATIONS WATERPROOF BAG

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

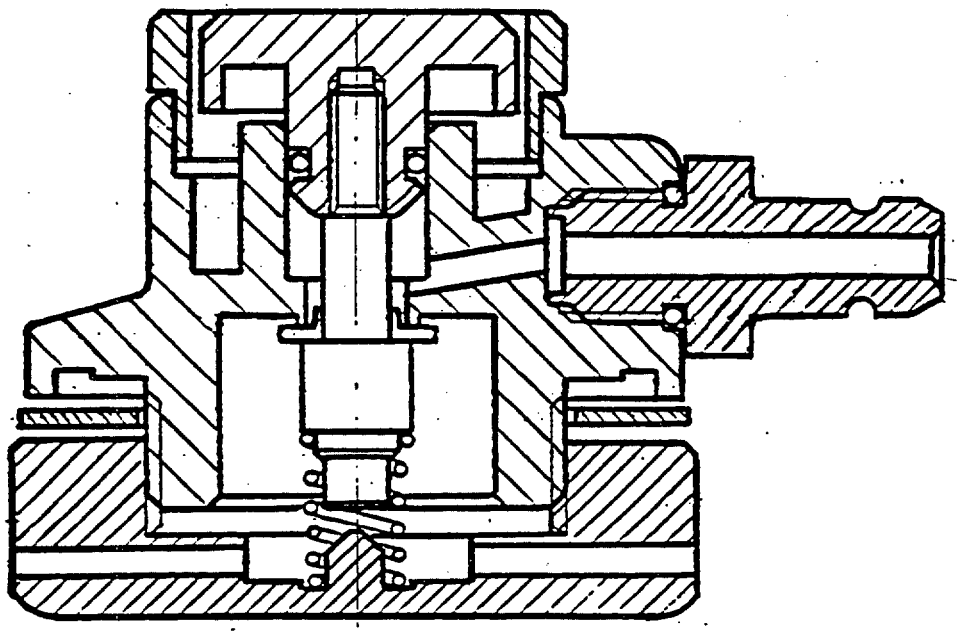


Fig 11

INLET VALVE

X-4815

SPECIAL OPERATIONS WATERPROOF BAG

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

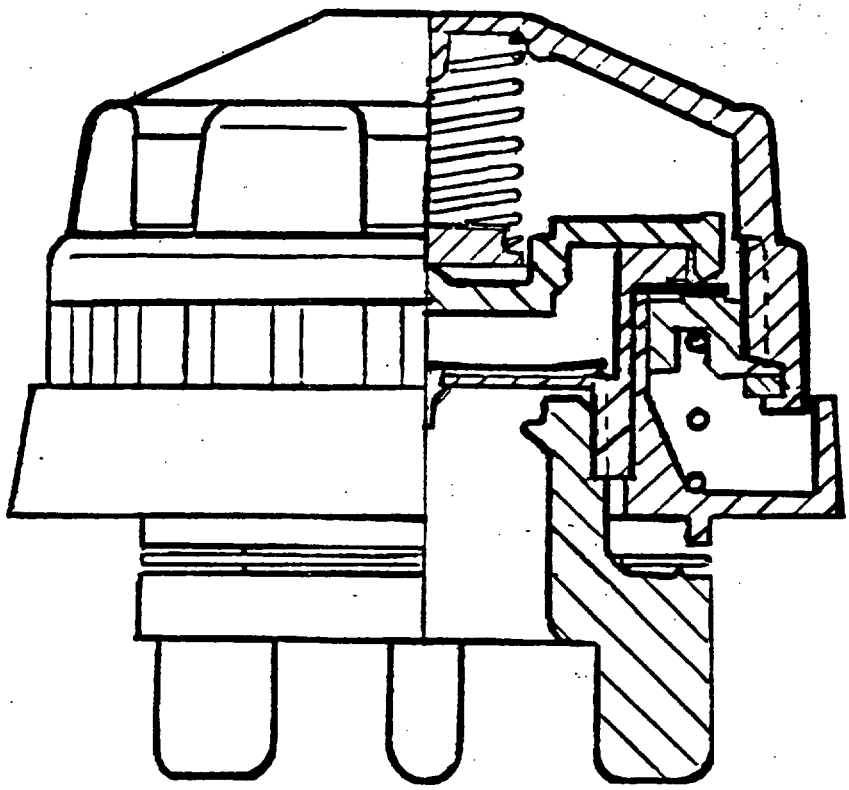


Fig 12

EXHAUST VALVE

X-4816

SPECIAL OPERATIONS WATERPROOF BAG

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

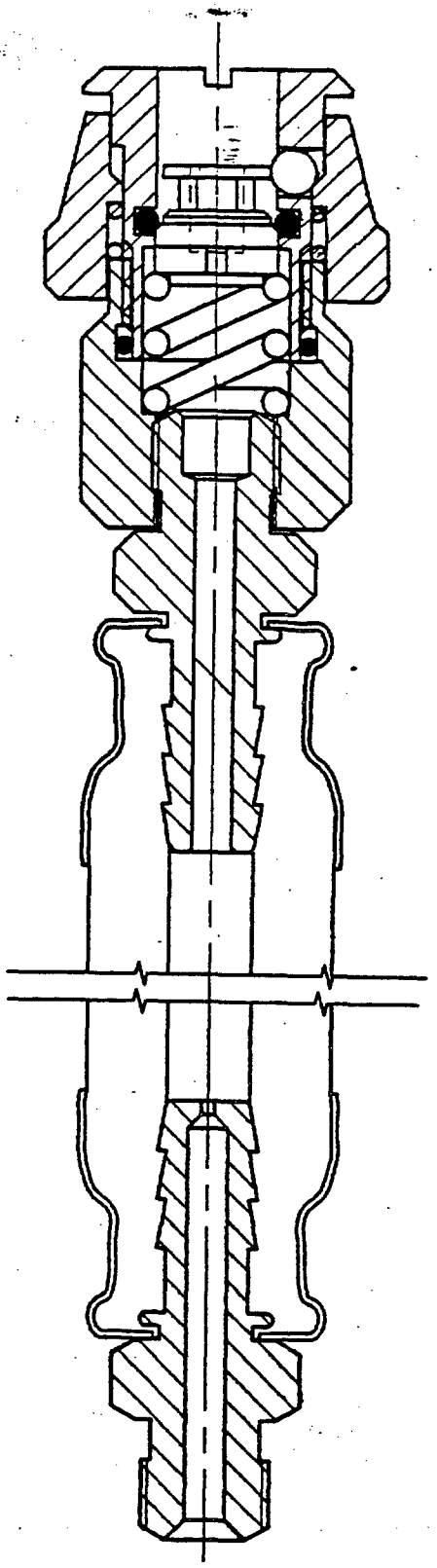


Fig 13

INFLATOR HOSE ASSEMBLY

X-4817

SPECIAL OPERATIONS WATERPROOF BAG

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

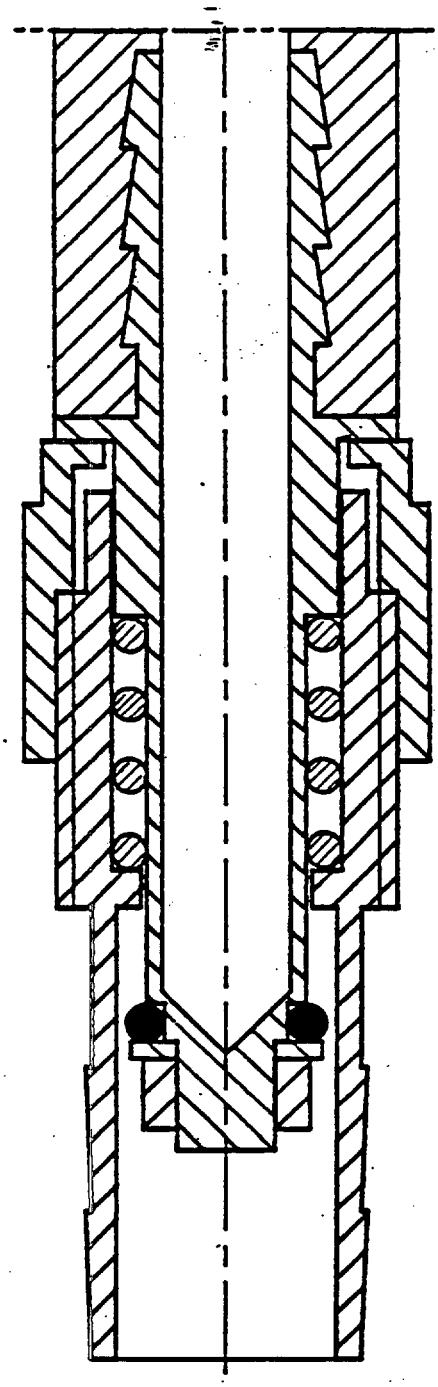


fig. 14

ORAL INFLATOR

X-4818

SPECIAL OPERATIONS WATERPROOF BAG

Source: https://assist.dla.mil -- Downloaded: 2026-10-26 04:12Z
Check the source to verify that this is the current version before use.